

L2 Introduction to Machine Learning Quiz

ANS

1. In supervised learning, what information does the teacher provide?

- A) Example inputs and correct outputs
- B) Only unlabeled input data
- C) Rewards without desired outputs
- D) A fixed policy with no examples

ANS: A) Example inputs and correct outputs

2. What is the primary goal of unsupervised learning?

- A) To predict only continuous values
- B) To discover structure or hidden patterns in unlabeled data
- C) To maximize cumulative rewards through interaction
- D) To compare predictions with teacher-provided labels

ANS: B) To discover structure or hidden patterns in unlabeled data

3. In reinforcement learning, what does an agent attempt to maximize?

- A) The number of input features
- B) The size of each mini-batch
- C) Its cumulative rewards
- D) The number of hidden layers

ANS: C) Its cumulative rewards

4. Which task is an example of regression?

- A) Classifying an email as spam or not spam
- B) Identifying an image as a cat or dog
- C) Predicting whether a statement is true or false
- D) Predicting a house price for next year

ANS: D) Predicting a house price for next year

5. Which statement correctly contrasts training and inference?

- A) Training adjusts model parameters through repeated forward and backward passes, whereas inference typically uses one forward pass.
- B) Training uses one forward pass, whereas inference requires millions of back-propagation iterations.
- C) Training always runs on an edge device, whereas inference requires a GPU cluster.
- D) Training and inference both update model weights in the same way.

ANS: A) Training adjusts model parameters through repeated forward and backward passes, whereas inference typically uses one forward pass.

6. Which type of value is normally predicted by a classification model?

- A) A continuous salary value
- B) A discrete class label
- C) A model's training time
- D) A learning-rate schedule

ANS: B) A discrete class label

7. What does a loss function measure in supervised learning?

- A) The number of layers in a neural network
- B) The size of the training dataset
- C) How much the predicted value differs from the ground-truth value
- D) The frequency of an edge processor

ANS: C) How much the predicted value differs from the ground-truth value

8. Which function is used in the lecture's logistic-regression model for binary classification?

- A) Mean absolute error
- B) K-fold averaging
- C) Random cropping
- D) Sigmoid function

ANS: D) Sigmoid function

9. What layers make up the two-layer fully connected neural network used to solve XOR in the presentation?

- A) One input layer, one hidden layer, and one output layer
- B) Two input layers and no output layer
- C) One convolution layer and one pooling layer only
- D) Five hidden layers and one input layer

ANS: A) One input layer, one hidden layer, and one output layer

10. What may happen if a neural network has too many layers and neurons?

- A) It must become a regression model.
- B) It may overfit the training data.
- C) Its model capacity always decreases.
- D) It can no longer use activation functions.

ANS: B) It may overfit the training data.

11. Which loss function is commonly associated with classification in the presentation?

- A) Mean Squared Error only
- B) Mean Absolute Error only
- C) Cross-Entropy Loss
- D) Quantile Loss only

ANS: C) Cross-Entropy Loss

12. Which loss function is commonly associated with regression?

- A) Focal Loss
- B) Hinge Loss
- C) Cross-Entropy Loss
- D) Mean Squared Error

ANS: D) Mean Squared Error

13. In multi-class image classification, what are the two main processing stages?

- A) Feature extraction followed by classification
- B) Back-propagation followed by data labeling
- C) Map construction followed by vehicle control
- D) Regression followed by reinforcement learning

ANS: A) Feature extraction followed by classification

14. What does the SoftMax output represent in a multi-class classifier?

- A) A list of gradients whose values sum to zero
- B) A list of class probabilities that sum to 1.0
- C) A list containing only ground-truth labels
- D) A list of hidden-layer sizes

ANS: B) A list of class probabilities that sum to 1.0

15. In a medical-test example, which outcome is a false positive?

- A) A sick person is diagnosed as sick.
- B) A healthy person is diagnosed as healthy.
- C) A healthy person is misdiagnosed as sick.
- D) A sick person is misdiagnosed as healthy.

ANS: C) A healthy person is misdiagnosed as sick.

16. In a medical-test example, which outcome is a false negative?

- A) A healthy person is correctly diagnosed as healthy.
- B) A healthy person is misdiagnosed as sick.
- C) A sick person is correctly diagnosed as sick.
- D) A sick person is misdiagnosed as healthy.

ANS: D) A sick person is misdiagnosed as healthy.

17. What is the purpose of back-propagation during neural-network training?

- A) To calculate gradients used to update model parameters
- B) To convert a regression problem into clustering
- C) To replace the forward pass during inference
- D) To divide data into cross-validation folds

ANS: A) To calculate gradients used to update model parameters

18. How does mini-batch stochastic gradient descent compute a parameter update?

- A) It always uses the entire dataset for every gradient.
- B) It uses a small portion of the training data to compute the gradient.
- C) It uses validation data as ground-truth test data.
- D) It updates parameters without calculating a loss.

ANS: B) It uses a small portion of the training data to compute the gradient.

19. What does a large gap between training accuracy and validation accuracy often imply?

- A) The model has no trainable parameters.
- B) The model is necessarily underfitting.
- C) The model may be overfitting.
- D) The validation set has become the training set.

ANS: C) The model may be overfitting.

20. In 5-fold cross-validation, how is each fold used?

- A) All five folds are used only for testing.
- B) One fold is permanently discarded.
- C) Each fold is used only for training and never for validation.
- D) Each fold serves as validation once while the other four serve as training data.

ANS: D) Each fold serves as validation once while the other four serve as training data.